

1 Package mathml

The `mathml` package is designed to convert a Maxima expression into Presentation MathML, version 2.0.

1.1 Definitions for package mathml

`mathml`

[Function]

```
mathml (expr)
mathml (expr, filename_or_false)
mathml (label, filename_or_false)
```

Convert an expression to Presentation MathML (version 2).

When the first argument is a label, the label and its associated value are printed verbatim, not typeset as MathML.

When the optional argument `filename_or_false` is supplied, any output will be appended to the file, unless it is `false`, in which case the output is returned as a string.

`load("mathml");` loads this package.

Examples:

```
(%i1) load(mathml) $
(%i2) mathml (integrate (sin(x), x));
      <math xmlns="http://www.w3.org/1998/Math/MathML"> <mo>-</mo>
      <mi>cos</mi> <mi>x</mi> </math>
(%o2)
      false
(%i3) integrate (1/(1 + x^3), x);
              2 x - 1
              atan()
              log(x^2 - x + 1)      sqrt(3)      log(x + 1)
(%o3)      - + +
              6          sqrt(3)          3
(%i4) mathml (%i3);
<pre>
(%i3) integrate(1/(x^3+1),x);
</pre>
(%o4)
      %i3
(%i5) mathml (%o3);
<pre>
(%o3) -(log(x^2-x+1)/6)+atan((2*x-1)/sqrt(3))/sqrt(3)+log(x+1)/3;
</pre>
(%o5)
      %o3
(%i6) mathml (``%o3);
      <math xmlns="http://www.w3.org/1998/Math/MathML"> <mo>-</mo>
      <mfrac><mrow><mi>log</mi> <mrow><mo>(</mo><msup><mrow><mi>x</mi>
      </mrow> <mn>2</mn> </msup> <mo>-</mo> <mi>x</mi> <mo>+</mo>
      <mn>1</mn> <mo>)</mo></mrow> </mrow> <mrow><mn>6</mn>
```

```

</mrow></mfrac> <mo>+</mo> <mfrac><mrow><mi>arctan</mi>
<mrow><mo>(</mo><mfrac><mrow><mn>2</mn>
<mspace width="0.16666667em"/><mi>x</mi> <mo>-</mo> <mn>1</mn>
</mrow> <mrow><msqrt><mn>3</mn> </msqrt></mrow></mfrac>
<mo>></mo></mrow> </mrow> <mrow><msqrt><mn>3</mn> </msqrt>
</mrow></mfrac> <mo>+</mo> <mfrac><mrow><mi>log</mi>
<mrow><mo>(</mo><mi>x</mi> <mo>+</mo> <mn>1</mn> <mo>></mo></mrow>
</mrow> <mrow><mn>3</mn> </mrow></mfrac> </math>
(%o6) false
(%i7) mathml (''%o3, "/tmp/tmp.xml");
(%o7) false
(%i8) mathml (''%o3, false);
(%o8)
<math xmlns="http://www.w3.org/1998/Math/MathML"> <mo>-</mo>
<mfrac><mrow><mi>log</mi> <mrow><mo>(</mo><msup><mrow><mi>x</mi>
</mrow> <mn>2</mn> </msup> <mo>-</mo> <mi>x</mi> <mo>+</mo>
<mn>1</mn> <mo>></mo></mrow> </mrow> <mrow><mn>6</mn>
</mrow></mfrac> <mo>+</mo> <mfrac><mrow><mi>arctan</mi>
<mrow><mo>(</mo><mfrac><mrow><mn>2</mn>
<mspace width="0.16666667em"/><mi>x</mi> <mo>-</mo> <mn>1</mn>
</mrow> <mrow><msqrt><mn>3</mn> </msqrt></mrow></mfrac>
<mo>></mo></mrow> </mrow> <mrow><msqrt><mn>3</mn> </msqrt>
</mrow></mfrac> <mo>+</mo> <mfrac><mrow><mi>log</mi>
<mrow><mo>(</mo><mi>x</mi> <mo>+</mo> <mn>1</mn> <mo>></mo></mrow>
</mrow> <mrow><mn>3</mn> </mrow></mfrac> </math>

```

The difference between the results of input line 6 versus 8 is the value of %o8 is bound to the string of MathML markup, while %o6 is false and the MathML markup has been printed to the terminal.

mathml_non_numeric_subscripts

[Option variable]

Default value: false

When **mathml_non_numeric_subscripts** is true and **mathml_underscore_is_subscript**, an underscore is treated as a subscript unconditionally. The default is that only substrings of digits are subscripts.

```

(%i1) load(mathml) $
(%i2) (mathml_non_numeric_subscripts:false, mathml_underscore_is_subscript:true)
(%i3) mathml(a_variable_name);
<math xmlns="http://www.w3.org/1998/Math/MathML">
<mi>a_variable_name</mi> </math>
(%o3) false
(%i4) mathml(a_variable_name_1_2);
<math xmlns="http://www.w3.org/1998/Math/MathML">
<msub><mi>a_variable_name</mi> <msub><mi> 1 </mi> <mi> 2 </mi> </msub></msub><
(%o4) false
(%i5) mathml_non_numeric_subscripts:true $
(%i6) mathml(a_variable_name);
<math xmlns="http://www.w3.org/1998/Math/MathML">

```

```

      <msub><mi>a</mi> <msub><mi> variable </mi> <mi> name </mi>    </msub></msub></mat
(%o6)
      false
(%i7) mathml(a_variable_name_1_2);
      <math xmlns="http://www.w3.org/1998/Math/MathML">
      <msub><mi>a</mi> <msub><mi> variable </mi> <msub><mi> name </mi> <msub><mi> 1 </

```

mathml_underscore_is_subscript

[Option variable]

Default value: false

When `mathml_non_numeric_subscripts` is true. In this case, when `mathml_underscore_is_subscript` is false, `mathml` treats an underscore as part of a symbol and only trailing digits in the symbol name are treated as subscripts; when true, all underscores in a symbol name are treated as if they begin a subscript; otherwise, when `mathml_underscore_is_subscript` is a positive integer n , the first n underscores in a symbol name (counting from the right) are treated as subscripts and any remaining subscripts are treated as ordinary characters.

When `mathml_non_numeric_subscripts` is false, the above description is accurate for the right-most part of the symbol name that contains only digits and underscores (compare the lines following %i4 and %i7 in the above example).

```

(%i1) load(mathml) $
(%i2) mathml_underscore_is_subscript:false $
(%i3) mathml(x_1);
      <math xmlns="http://www.w3.org/1998/Math/MathML">
      <msub><mi>x</mi> <mn>1</mn></msub> </math>
(%o3)
      false
(%i4) mathml_underscore_is_subscript:1 $
(%i5) mathml(x_1);
      <math xmlns="http://www.w3.org/1998/Math/MathML">
      <msub><mi>x</mi> <mi> 1 </mi>    </msub></math>
(%o5)
      false
(%i6) mathml(x_1_2);
      <math xmlns="http://www.w3.org/1998/Math/MathML">
      <msub><mi>x_1</mi> <mi> 2 </mi>    </msub></math>

```

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